Zahra Khodagholi

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Summary — Ph.D. Student in Industrial Engineering at the University of Central Florida (expected 2027), specialising in artificial intelligence, machine learning, medical imaging and generative models. Experienced in research and development of spatiotemporal networks, uncertainty calibration, and computational biology. Seeking opportunities to leverage research expertise and strong software engineering skills to build impactful ML products.

Skills

Programming Python, C++

Frameworks PyTorch, TensorFlow, Keras, OpenCV, NumPy, Pandas, Git

Algorithms Linear Regression, Logistic Regression, K-means, K-nearest Neighbour, Random Forest, CNNs

Education

University of Central Florida

Ph.D. in Industrial Engineering — *May 2025 – Present (Expected September 2027)*

- GPA: 3.77/4.00
- Research Interests: Artificial Intelligence, Computational Biology, LLM Foundation Models, Generative AI.

Master of Computer Science — *Sep 2022 – May 2025*

- GPA: 3.70/4.00
- Focus areas: Computer Vision, Machine Learning, Medical Imaging.

Al-Zahra (Azzahra) University

Bachelor of Science in Computer Engineering — *Sep 2016 – Aug 2020*

- GPA: 16.18/20.00 (Last 60 credits: 17.17/20.00)
- Thesis: Object Detection using ResNet.

Publications

From Rules to Foundation Models: A Comprehensive Review of Machine Learning Approaches for siRNA Design **2026**

NAR Genomics and Bioinformatics, 8(3), lqag099

- Surveyed 40+ studies spanning two decades of siRNA efficacy and off-target modeling, from classical design rules to transformers, graph neural networks, and RNA foundation models. Contributed a method taxonomy, formal saliency validation protocols, and recommendations for benchmark standardization (advisor: Niloofar Yousefi).
- DOI: 10.1093/nargab/lqag099

Validating Interpretability in siRNA Efficacy Prediction: A Perturbation-Based, Dataset-Aware Protocol **2026**

ICLR 2026 Workshop on Machine Learning for Genomics Explorations (MLGenX)

- Introduced a perturbation-based faithfulness protocol that validates gradient-based saliency maps for siRNA sequence design. Identified two novel cross-dataset failure modes (faithful-but-wrong and inverted saliency) and proposed BioPrior, a biology-informed regularizer that strengthens explanation faithfulness across four benchmarks (advisor: Niloofar Yousefi).
- arXiv — Code

ULTRA-AIR: Ultrasound Landmark Tracking for Real-time Anatomical Airway Identification and Reliability Check **2024**

IEEE International Conference on Body Sensor Networks

- Added epistemic uncertainty to YOLOv9 for airway landmark detection, yielding calibrated confidence with standard bounding boxes.
- IEEE Xplore — Code

Concept-Conditioned Message Passing: Interpretable siRNA Efficacy Prediction with Provable Intervention Isolation 2026

Submitted to NeurIPS 2026

- Proposed CCMP, a FiLM-modulated heterogeneous SAGE GNN that provably achieves per-concept intervention isolation for interpretable siRNA efficacy prediction. Validated zero-shot recovery across all 8 FDA-approved siRNA therapeutics and released leakage-audited cross-dataset benchmarks (advisor: Niloofar Yousefi).

Experience

Brainchip **Jun 2025 – Sep 2025**
Machine Learning Intern
- Built and deployed an INT8-quantized spatio-temporal fall-detection model on BrainChip's Akida using TNP-B/TENNs-B. Added BatchNorm, temporal smoothing (optimal decay 0.85–0.90) and a 16-frame FIFO buffer to overcome dataset bias and gradient collapse, achieving **98.36%** accuracy (+0.13% vs. Float32) with ultra-low power consumption.
- Designed a robust data pipeline for the FallVision dataset (11,732 videos; 6,002 falls): automated archive extraction, corrupt video filtering, normalization to 10 FPS at 224×224, category-aware balancing, TFRecord conversion and thread-pooled parallel processing.
- Validated the INT8 model on Akida hardware and delivered a complete solution, including preprocessing scripts, performance benchmarks and technical documentation; collaborated with mentors and produced a FallVision – BrainChip Fall Detection System presentation summarising the results.

University of Central Florida **Aug 2023 – Dec 2024**
Graduate Teaching Assistant
- Conducted Java programming sessions and graded exams and assignments for Computer Science II and System Administration courses.

ToobaTech Company **Aug 2019 – Jul 2020**
Machine Learning Intern
- Applied and implemented machine learning algorithms, including Linear Regression, Logistic Regression, K-means clustering, K-nearest Neighbour and Random Forest, to support business insights and product prototypes.

Boston University **Jul 2021 – Dec 2021**
Computer Vision Intern
- Utilised computer vision techniques to segment paediatric lung images for COVID-19 diagnosis; completed the image segmentation component and contributed to editing chapters of a machine learning textbook.

Research Experience

Breast Cancer Segmentation **Mar 2024 – May 2024**
University of Central Florida
- Established a baseline UNET model using MONAI for breast cancer detection and segmentation on the TIGER dataset (advisor: Chen Chen).

Branch Prediction Using CNNs **Feb 2023 – Apr 2023**
University of Central Florida
- Designed a branch predictor using convolutional neural networks, improving prediction accuracy; implemented Gshare, bimodal, Smith N-bit counter and hybrid predictors on provided traces (advisor: Jongouk Choi).

COVID-19 Diagnosis in Children **Jul 2021 – Dec 2021**
Boston University
- Utilised computer vision techniques for lung image segmentation to aid COVID-19 diagnosis in children; completed the segmentation component of the project (advisor: Reza Rawassizadeh).

Using VR/AR and Artificial Intelligence in the Judicial System **Mar 2019 – Jun 2019**
Alzahra University
- Conducted a literature review on VR/AR and AI technologies for potential applications in the judicial system; see study (advisor: Masoud Sagharichian).

Projects

Object Detection using Retinanet **Sep 2019 – Jul 2020**
Toobatech Company / Alzahra University
- Bachelor's thesis: trained the COCO dataset on a RetinaNet architecture using a ResNet backbone to perform object detection.

Web Browser from Scratch **May 2019 – Jul 2019**
Alzahra University
- Implemented the TCP/IP process from scratch in Python without using built-in libraries; received a course grade of 18/20.

Implementation of AI Algorithms **May 2019 – Jul 2019**
Alzahra University
- Implemented classical AI algorithms (A* search, constraint satisfaction problem solvers, best-first search and hill climbing) in Python.

Honors & Awards

- Ranked among the top 1.5% in Iran's national university entrance exam (Konkour) out of more than 200,000 students.
- Awarded a full tuition-fee waiver for the Bachelor of Science degree.